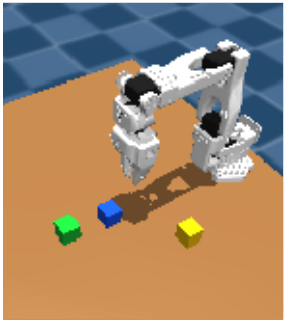
(a) RGB-D observation

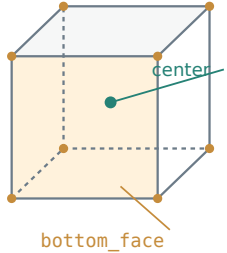


Recorded side RGB
+ calibrated depth

(b) Metric geometry

Points, edges, and faces

Schematic geometry



Geometry + evidence
Hidden surfaces are inferred

(c) Text + symbols

object: green

center [m]
(0.12, -0.01, 0.10)

goal: place on blue

to_goal [m]
(-0.02, 0.03, -0.04)

Illustrative values; world frame

green.bottom_face

blue.top_face

(d) Agentic policy

LLM reasoning
Generate / revise a policy
Between training rollouts

Policy execution
Local sensing + pick / place

Prior policy experience
Planned online reuse

Planned online loop: reason, execute, observe — no reset until success or timeout

Gray dashed: hidden geometry Purple dashed: planned online use